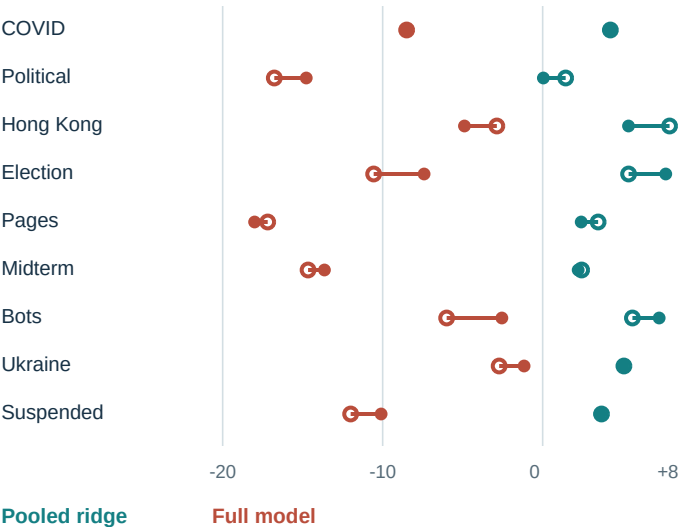


# Better pooled features, worse transfer

**Finding:** on bot detection, continued graph-ICL pretraining improves information accessible to a fixed readout before the learned projection, yet degrades end-to-end predictions across nine sources. The same training improves three other targets. This is a target-dependent divergence, not universally harmful training.

## A Opposite trajectories across nine sources

TwiBot20: within-episode AUC change, step100 to2500



Filled/open: original/fresh streams; 128 episodes per cell, one training seed per source. All18 signs agree in each readout. Mean: pooled +4.21/+4.36 points, full -9.00/-10.14. Signs are not individual significance claims.

## B The counterexample is the same training

Full-model pooled AUC change; source mean, points

| Target    | Original | Fresh  |
|-----------|----------|--------|
| Political | +21.70   | +20.82 |
| Election  | +36.53   | +37.79 |
| Pages     | +15.80   | +15.62 |
| Bots      | -8.18    | -9.30  |
| Suspended | +0.07    | -1.13  |

All nine full models improve on political, election and pages in both streams; all nine decline on Bots. Source breadth does not equal target breadth. These are endpoint changes, not monotonic trajectories.

**Location matters:** the improving probe is S0\_pool, before reset\_mlp\_m. U1 ridge usually declines too. The evidence does not isolate the metagraph solver, and the entire encoder cannot be called improving.

## C What the interventions explain

**Not just an abrupt swap:** freezing the initial readout during18 matched trainings improves the Facebook probe in all nine source-seed pairs on both streams, but fails the full-model repair criterion. The probe improvement is a secondary result, not success on the declared primary.

**Not a generic cure:** the completed eight-arm gradient-isolation study loses to native training on fixed-panel AUC under both schedules. Native training can help representations even when a fixed readout performs better at inference.

## D What remains unexplained

Matched page episodes show that support-context preference can reverse only for native inference, not for a fixed prototype. Political interventions locate a support-to-reference path with queries unchanged. These explain why ablation signs cannot be read as encoder quality; they do not explain the nine-source bot trajectory causally.

**Public boundary:** a useful original-style Wiki-to-FB15K-237 checkpoint reaches73.8% accuracy; suppressing supports or queries drops it to39.9%/39.0%. Both roles can be essential. No universal support-harm claim.

**Contribution candidate:** a source-broad, target-dependent divergence between improving pooled readout performance and worsening end-to-end graph transfer, supported by matched trajectories and training interventions. The practical consequence is to evaluate both stages: a single full-model score does not identify whether pretraining improved the underlying pooled features.

**Decision:** build a phenomenon paper, not a claimed repair or predictive mechanism. Keep the source/target boundaries visible; do not expand the failed isolation method. A graph-specific explanation that predicts the affected targets is still missing.

Evidence: corrected sampler replay (311383ab; nine runs, four steps,6120 rows); frozen-readout training (f20e6495; 18 models, three seeds); isolation (0b9d9641); matched page (8ccd3686); public quality/roles. Public replay used an explicit accuracy-equivalence amendment after the original logit check failed; no fixed-query mediation claim for that setting.